BioSynvia | Product Requirements Document

Reproducible Bioprocess Soft-Sensing & Hybrid Modeling Workbench

Independent Portfolio Research Project
November 4, 2026

Problem Statement — What BioSynvia Solves

The problem. CHO cell-culture reactors generate frequent sensor measurements, but important biological attributes may only be measured through intermittent laboratory assays. This creates gaps in direct visibility between assay times. Predictive models could help, yet results may be misleading if future information leaks into features or if closely related reactor runs appear in both training and testing.

Research question. Can available inline sensor data estimate a verified, sparsely measured biological attribute reliably on entirely unseen CHO cultivation experiments?

Proposed solution. BioSynvia will compare statistical and machine-learning soft sensors using experiment-group-held-out evaluation, quantify error and limitations, and display qualified predictions through a read-only scientific dashboard. Mechanistic/hybrid methods remain conditional on data suitability.

Expected outcome. A reproducible research prototype with a documented dataset, defensible model comparisons, independent test results, uncertainty or failure analysis, and transparent interpretation—not a validated manufacturing Digital Twin or process-control system.

Document map

Part	Where to look
01 — Scientific foundation	Sections 1–5: vision, naming/data rights, CHO dataset, users, scope
02 — Product and modeling design	Sections 6–9: requirements, statistical/hybrid experiments, UX, architecture
03 — Execution and assurance	Sections 10–14: quality gates, release decisions, roadmap, milestones, risks
04 — Delivery and research evidence	Sections 15–18: portfolio package, unresolved decisions, first sprint, sources

Tip: in Word, use the Navigation pane to jump directly between numbered sections.

Document control

Field	Detail
Product / working name	**BioSynvia**
Descriptive subtitle	**Reproducible Bioprocess Soft-Sensing & Hybrid Modeling Workbench**
PRD version / date	1.1 / November 4, 2026
Product owner / developer	Lakshmi Sindhu Pulugundla (proposed individual portfolio build)
Primary audience	Project owner; scientific modeling reviewer; engineering collaborator; hiring-manager demo audience
Product type	Offline, research-only scientific modeling prototype
Core dataset	Uhlendorff et al., CHO-K1 cultivation dataset, Zenodo DOI **10.5281/zenodo.20829178** [S1]
Target environment	Local laptop; Python; reproducible Git repository; optional local Streamlit dashboard
Project state at PRD issue	**Planned.** No BioSynvia model results, measured improvements, or working product are claimed.
Naming status	**Provisional / elevated trademark-similarity risk.** See §2.
Approval gate	Technical scope approved for internal development only; public branding and data redistribution require separate clearance.

> **Evidence policy.** Throughout this PRD, **VERIFIED SOURCE FACT** means explicitly supported by a cited dataset or official reference. **PROPOSED** means an engineering requirement, experiment, hypothesis, or design decision. **GATED** means implementation depends on a data-access, biological, statistical, or legal check. No hypothetical performance is represented as observed.

1. Executive summary

1.1 Product vision

BioSynvia is a reproducible workbench for estimating sparsely measured bioprocess attributes from frequently sampled cell-culture sensors, evaluating model reliability on new cultivation experiments, and helping scientific users inspect predictions and their limitations. The initial application is Chinese hamster ovary (CHO-K1) cultivation at laboratory bioreactor scale; it is **not** directly a CAR-T, AAV, or clinical-development system.

Its flagship output is not an attractive accuracy number. It is a **complete scientific decision trail**: raw-data provenance, justified target selection, training/test discipline, baselines, model comparison, group-held-out performance, uncertainty or failure analyses, and transparent end-user presentation. Later stages may extend the statistical soft sensor with a simplified mechanistic or hybrid model **only if the available variables support identification and evaluation**.

1.2 Detailed problem statement and opportunity

Cell-culture facilities often have frequent process-sensor measurements but relatively infrequent laboratory assays. A scientist might want a time-resolved estimate of a biological state such as viable cell concentration, but direct measurement may be sparse, delayed, expensive, or otherwise intermittent. A model can potentially infer an attribute from available sensor signals. However, a model that performs well on randomly mixed records from the same culture may fail on a new experiment, process mode, or sensor configuration. This undermines trust, interpretability, and usefulness.

BioSynvia focuses on an under-addressed engineering requirement in many portfolio demonstrations: **evaluation must represent the scientific unit of independence**. Here, 24 reactor runs arise from 12 experiments with paired reactors. Both reactors from an experiment must stay in the same data partition. [S1, S4]

1.3 Target outcome

- A reproducible pipeline from source archive to documented, model-ready data.
- A defensible **offline soft-sensor evaluation** on a measured target selected after the data audit.
- A comparison of simple, statistical, chemometric, and selected nonlinear methods.
- Group-held-out evaluation, per-experiment error reports, failure diagnostics, and appropriately qualified uncertainty.
- A scientist-facing local dashboard with provenance, measured-versus-predicted trajectories, model status, and limitations.
- An optional mechanistic/hybrid modeling extension with explicit eligibility and scientific validity gates.
- A publishable code-and-report portfolio **without implying process validation, clinical utility, regulatory compliance, or an operating Digital Twin**.

1.4 Explicit boundaries

This product is a research prototype. It will **not** automate process control, release manufacturing batches, recommend clinical treatment, claim GxP validation, connect to live bioreactors, prove cell/gene therapy expertise, or count as professional biotech employment. Its purpose is to demonstrate scientific modeling reasoning with reproducible evidence.

2. Working name, legal/ethical boundaries and publication strategy

2.1 BioSynvia name decision — elevated risk

BioSynvia is approved by the project owner as an **internal, provisional project name**, not a cleared trademark. A third party uses **SYNVIA** commercially in pharmaceutical research, and a U.S. SYNVIA registration is reported for scientific research and software-related services (registration **7505668**, serial **98083883**). Another SYNVIA record exists for medical services; additional newer applications also surface. The business/domain overlap is material. The USPTO explains that marks may conflict due to similarity in sound/appearance and related goods or services even if the names are not identical. [S5–S7]

Branding gate BR-01: Before launching a public product, paid offering, domain, social account, logo or brand-focused portfolio site, obtain professional trademark advice or conduct and document a more complete rights review. Until then, use the phrase **“BioSynvia (working title) — independent academic-style research prototype”**, refrain from suggesting affiliation with SYNVIA or Johnson & Johnson, and keep the repository name changeable. A PRD is not legal clearance.

2.2 Dataset copyright and redistribution

The Zenodo dataset page lists the archive as publicly accessible but **does not show a populated dataset-license value in its Rights panel** at this review date. A related November 2026 preprint says the *article* is offered under CC BY 4.0, but that is **not sufficient evidence that the underlying 227.6 MB archive is licensed for republication**. [S1–S2]

Data gate DG-01: Until explicit archive reuse rights are confirmed, the public repository should contain code, schema, documentation, small synthetic test fixtures, dataset citations and instructions to download from the original source; it should **not** re-host the original measurement files. Preserve author attribution, provenance, published DOI, archive checksum, and any author-supplied conditions. Seek permission if planned use exceeds confirmed rights.

2.3 Research ethics and transparency

All model cards and dashboards must display: *Research prototype; retrospective analysis; not validated for process control, manufacturing decisions, clinical use or GMP/GxP release.* Disclose which analyses used actual laboratory records versus generated synthetic data. All figures must distinguish measured, fitted, interpolated, and extrapolated values; no fabricated performance figures may appear in demos or application materials.

3. Scientific context and dataset specification

3.1 Primary source — verified description

The source is **“Dataset Based on Chinese Hamster Ovary (CHO) Cultivations including Turbidity, Permittivity, O2 and CO2 Measurements”** by Selina Uhlendorff, Ruwen Fulek, Jonas Eimler, Miriam Pein-Hackelbusch, and Björn Frahm. It was published on Zenodo on **August 11, 2026** under DOI **10.5281/zenodo.20829178**. The main dataset archive is listed at **227.6 MB**, with MD5 checksum **71b2a35b7df2ed11644ec9f177453772**. [S1]

Dimension	Verified record	Implementation consequence
Cell line	CHO-K1 mammalian cells	Domain is biologics upstream cell culture, not CGT manufacturing
Cultivations	**24** reactor cultivations	Use reactor-level diagnostics, not 24 assumed-independent treatment groups
Experimental grouping	**12** experiments, each with 2 parallel reactors	Split by experiment ID; paired reactors never straddle partitions
Process modes	**9 batch**, **3 fed-batch** experiments	Report mode-stratified results; fed-batch conclusions will have high uncertainty
Duration	**161.5–328.5 hours**	Unequal duration, irregular missingness, phase sensitivity
Reactor working volumes	**0.55–0.8 L**	Laboratory-scale scope; no demonstrated industrial scale-up
Variables	**48 total: 38 inline + 10 offline**	Potential predictor/target separation, to be verified per actual column
Offline sampling	Up to twice daily	Targets available at sparse assay times; no pseudo-label inflation
Raw files	**12 Excel workbooks**, 16 worksheets per workbook	Raw ingest needs worksheet inventory and time-axis reconciliation

Dimension	Verified record	Implementation consequence
Aligned files	**48 CSVs**, inline + offline for each reactor	Initial parsing accelerated but alignment provenance must be audited
Sensor modalities	Includes turbidity, permittivity, O2 and CO2 measurements	Candidate signal families; exact units and features pending inspection

Important: the above describes the repository record. The archive was **not downloaded and profiled as part of authoring this PRD**. Exact column names, assay methods, units, target completeness, reactor metadata, alignment algorithms, and missingness rates are **open Phase 1 questions**, not verified implementation details. [S1]

3.2 Physical and statistical meaning

Inline measurements are sampled continuously or frequently using installed instrumentation, potentially providing real-time insight into culture conditions. **Offline** variables are laboratory assays collected much less frequently and may better reflect biological attributes that sensors do not measure directly. A soft sensor is a computational estimate of one of those hard-to-observe attributes based on signals available at inference time.

Sampling poses three distinct risks:

1. **Temporal alignment:** Sensor clocks may drift; the authors provide aligned tables. Verify how values were aligned, whether interpolated values are flagged, and the inference-time interpretation.
2. **Target sparsity:** An offline result exists only when assayed. Evaluation at artificially interpolated target points would inflate sample size and create false confidence.
3. **Experimental dependence:** The paired reactors share experiment design. Random rows, or random reactor splits that separate a pair, would leak closely related experimental context into evaluation.

3.3 Proposed entity/schema model

Entity	Primary keys / fields	Purpose
experiment	experiment_id, process mode, provenance/source file, design metadata	Scientific grouping and split assignment
reactor_run	(experiment_id, reactor_id), start/end, source checksum	Unit for trajectories and reactor-level diagnostics
sensor_sample	run ID, elapsed hours, sensor name, value, unit, quality flag, source timestamp	Long-form inline observations
offline_assay	run ID, elapsed hours, assay type, result, unit, assay quality/status	Ground-truth targets at observed sample times
feature_snapshot	run ID, cutoff timestamp, feature version, values, missing flags	Only information available by prediction time
prediction	snapshot ID, model version, prediction, optional interval, applicability flags	Auditable model output
experiment_result	split, model, metric, error, subgroup, counts, seed, commit hash	Reproducible evaluation and comparison

Time rules: Store raw, aligned, and feature-construction timestamps separately. Normalize units only with documented conversion rules; do not silently treat absent units as interchangeable. Ensure sensor features use data at

or before the prediction timestamp. If laboratory reporting delay is known, any prior offline-assay feature must satisfy the availability timestamp rather than the nominal collection timestamp.

3.4 Target selection gate (research decision T-01)

The preferred candidate is **viable cell concentration/density**, but this is **not confirmed as a usable column**. Select the first target satisfying all of the following after Phase 1:

- A named, experimentally measured offline field with documented physical meaning and unit.
- Nontrivial coverage across multiple *independent* experiment IDs and both major modes where possible.
- Enough measured time points to support grouped evaluation without inventing labels.
- An unambiguous prediction use case and permissible observed-at-time covariates.
- No direct leakage from post-assay fields or duplicate target-derived sensors.

If viable cell density does not satisfy these criteria, examine another measured offline attribute (for example, a documented analyte), or **stop and narrow the research question**. Never choose a target merely because it produces impressive metrics.

3.5 Data quality and availability outputs

Phase 1 must produce dataset_inventory.csv, field_dictionary.csv, unit_conversion_log.csv, missingness_by_run.csv, timestamp_alignment_report.md, source_manifest.json (DOI, URL, checksums, licenses), target_selection_decision.md, and one representative trajectory comparison for a batch and fed-batch experiment.

4. Target users and decision scenarios

Persona	Need	Prototype interaction	Success condition
Bioprocess scientist	Understand process-state trajectories and when signals are unreliable	Select a culture/run; compare measured and estimated target; inspect suspicious periods	Can identify which points are measured, estimated, or unsupported
Applied modeler / scientist-engineer	Assess experimental assumptions, feature provenance and generalization	Compare models, experiment-held-out errors, calibration/uncertainty, model cards	Can reproduce a model ranking and explain why it was selected
Data/automation engineer	Understand input contracts, data provenance, time alignment and model artifacts	Inspect schemas, data quality report, input/output manifest	Can recreate inputs and identify missing/invalid signals
Technical hiring reviewer	Judge modeling depth and scientific caution	Review 6-slide case study and demo with failed cases and limitations	Can audit the claims without trusting a polished UI

Primary jobs to be done

- *When I have frequent sensor readings and sparse assay results, I want a qualified estimate of a selected biological attribute so I can investigate temporal changes.*
- *When a model seems accurate, I want to know whether it generalizes to an unseen experiment and which operating conditions are out of scope.*

- *When a prediction is uncertain or based on invalid inputs, I want a visible warning instead of a confident answer.*

5. Scope, releases and non-goals

5.1 MVP / V1 — required

1. Source manifest, license status and reproducible ingestion from legally obtained files.
2. Audit-ready canonical data model, target decision, unit and time contracts.
3. Group-aware training/development and frozen test selection.
4. Naive and statistical baselines plus at least one nonlinear candidate, where appropriate.
5. Final held-out metrics and per-experiment diagnostics with failed cases retained.
6. Honest uncertainty/applicability reporting, even if calibrated intervals are infeasible.
7. Read-only Streamlit scientific dashboard and machine-readable experiment outputs.
8. Readme, runnable reproduction steps, model card, tests and concise hiring-manager presentation.

5.2 V1.1 / advanced — conditional

- A simplified mechanistic ODE model supported by available substrates, feed/volume and biomass-related variables.
- Parameter estimation, identifiability checks, numerical verification and sensitivity analysis.
- Data-driven correction / hybrid model benchmarked against the same held-out experiments.
- Optional scenario analysis that clearly separates measured-data support from counterfactual assumptions.
- Broader data modality (CHO IR/fluorescence) **only as a separately attributed and separately validated extension**, not as if the datasets came from the same experiments.

5.3 Explicit non-goals

- Production Digital Twin, online industrial integration, closed-loop control, GMP/GxP validation or validation for process release.
- Clinical, patient or treatment recommendations; CAR-T target discovery; therapeutic claims.
- Claims of manufacturing process optimization or business savings not measured against real use.
- Deep learning for prestige; automatic integration with proprietary or J&J systems.
- Re-hosting uncertain-license raw datasets; use of data without documented rights.
- Self-promotional “90–95% fit” scores as substitutes for demonstrated competence.

6. Functional product requirements

Each requirement has a stable ID, priority, proposed behavior and **testable acceptance criterion**. The MVP cannot pass if a critical requirement fails.

ID	Priority	Requirement	Acceptance test / evidence
FR-01	P0	Archive provenance and integrity	Source DOI, expected MD5, file manifest and retrieval timestamp recorded; checksum mismatch fails ingestion
FR-02	P0	Hierarchical file inventory	All expected

ID	Priority	Requirement	Acceptance test / evidence
			experiment/reactor/measurement combinations reconciled; missing or duplicate files trigger explicit flags
FR-03	P0	Typed, unit-aware canonical data	Every used feature and target has type, unit, valid-range policy, missingness flag, source field
FR-04	P0	Time-alignment audit	Original/aligned timestamps and alignment assumptions documented; join tolerances tested against known fixtures
FR-05	P0	Target selection record	Signed-off target name, assay meaning, coverage and no-leakage feature rules in a versioned decision log
FR-06	P0	Experiment-group split engine	A unit test proves no experiment ID, including either paired reactor, appears on both sides of any split
FR-07	P0	Past-only feature generation	Automated tests inject future sensor perturbations and confirm earlier predictions remain unchanged
FR-08	P0	Baselines and candidate models	Training-only preprocessing; naive baseline plus at least Ridge/PLS and one nonlinear candidate where feasible
FR-09	P0	Reproducible evaluation	Frozen fold IDs, seeds, code/config versions, target units and run counts saved with every metric
FR-10	P0	Scientifically honest report	MAE/RMSE and baseline deltas plus per-experiment errors and limitations; failed hypotheses retained
FR-11	P0	User-facing visual review	Dashboard distinguishes measured vs predicted vs missing; error and coverage views accessible
FR-12	P0	No unsupported deployment claims	README, model card and UI visibly label research-only, offline and non-validated status
FR-13	P1	Applicability and uncertainty	Warning when missing key sensors/out-of-distribution;

ID	Priority	Requirement	Acceptance test / evidence
			interval coverage shown if statistically justified
FR-14	P1	Failure analysis and ablations	At least sensor-group ablation, process-mode assessment and experiment-level failure table
FR-15	P1	Model experiment registry	Models, pipeline objects, split manifest, config, metrics and environment lock packaged and load-tested
FR-16	P2	Mechanistic simulation	ODE solver tests conservation/sign conventions or documented deviations; assumptions available for review
FR-17	P2	Hybrid model benchmark	Same split and metrics as V1 baselines; clearly states whether hybrid helps or harms
FR-18	P2	Scenario explorer	Any scenario view labeled hypothetical; bounds restricted to supported/declared experimental settings

Priorities: **P0** = required to present as credible V1; **P1** = important robustness; **P2** = gated advanced extension.

7. Scientific methodology and experiment plan

7.1 Intended prediction task

Primary task (PROPOSED): Given available inline sensor measurements up to time *t*, estimate the chosen offline target at time *t* in an experimental reactor that was **not used to fit the model**. The target is scored only at actual measured offline timestamps.

Clearly distinguish:

- **Sensor-only inference:** no historical offline target values are available at runtime. This is the default primary soft-sensor contract.
- **Lab-assisted inference:** the most recent previously reported offline measurement can be used, only if its availability time is before *t*. Evaluate separately; never compare to sensor-only models without acknowledging different inputs.
- **Forecasting:** predicting future values beyond *t* is a different task, with additional leakage and horizon controls. It is out of MVP unless explicitly approved as an extension.

7.2 Unit of generalization and holdout design

- **Group key:** experiment_id, *not* row or reactor ID; keep the two parallel reactors together.

- **Initial proposed final holdout:** reserve two complete experiments (ideally one batch and one fed-batch **if suitable target coverage permits**). Freeze IDs before inspecting final test performance.
- **Model development:** use the remaining ten experiment groups for GroupKFold, LeaveOneGroupOut or another preregistered group-safe strategy; tune within development groups only.
- **Stratification caveat:** with just three fed-batch experiments, held-out performance estimates by mode are unstable. Report this explicitly and avoid sweeping process-mode claims.
- **If no valid balanced two-group holdout exists:** use grouped leave-one-experiment-out evaluation with nested tuning; designate no single inflated "unseen final test" score and disclose the limitation.
- **Temporal consistency:** if validation involves a within-run prediction setting, all features must remain past-only. Group independence does not by itself prevent within-row future-time leakage.

External reference: scikit-learn documents LeaveOneGroupOut/GroupKFold as methods ensuring that samples from the same group do not cross folds. [S4]

7.3 Preprocessing and feature-policy checklist

- Parse source time offsets and aligned tables; document interpolation, smoothing and unit handling.
- Profile outliers, sensor dropouts, flatlines, spikes, duplicate time points and negative/impossible readings.
- Define a *maximum tolerated alignment lag* from observed sampling behavior; do not silently match widely separated measurements.
- Create past-only rolling medians/means, slopes, variance, missingness and elapsed-time features with a versioned configuration.
- Treat experiment-level design settings as features only if they would genuinely be known at inference and do not encode target-dependent outcomes.
- Fit imputation, scaling, PCA/PLS and feature selection **inside training folds**; never fit on all records before splitting.
- Keep a feature_eligibility.csv table stating source, unit, timestamp availability, intended interpretation and explicit leak risk.

7.4 Candidate methods and rationale

Family	Candidate	Scientific benefit	Failure mode to investigate
Naive	Training-set median, last valid observed assay (separate scenario)	Checks whether modeling adds value	Trivial apparent gains from different information contracts
Single-sensor	Calibration of one physiologically plausible inline measurement	Transparent reference model	Sensor drift/nonlinear response
Penalized regression	Ridge / Elastic Net	Interpretable, stable with limited independent groups	Nonlinearity, collinearity, coefficient instability
Chemometric	Partial least squares (PLS) regression	Designed for correlated multivariate predictors	Component over-selection and optimistic preprocessing
Nonlinear trees	Random Forest / histogram gradient boosting	Captures interactions without neural-network overhead	Extrapolation and unstable generalization
Probabilistic	Small-feature Gaussian process	Predictive uncertainty candidates	Miscalibration and poor scaling

Family	Candidate	Scientific benefit	Failure mode to investigate
Mechanistic (gated)	Simplified substrate/biomass ODEs	Scientifically interpretable dynamics	Missing state variables, wrong equations, unidentifiable parameters
Hybrid (gated)	ODE prediction + regularized residual model	Tests added value of process assumptions	Error propagation and overfitting tiny group counts

Neural networks, Transformers, LSTMs and digital-twin stacks are **not** MVP requirements. They may be explored only if a data-supported hypothesis and group-held-out performance justify them.

7.5 Model selection protocol

1. Choose target and inference-time input contract, then **freeze** split groups and primary metric.
2. Establish naive and transparent sensor baselines before expensive modeling.
3. Run grouped cross-validation for each candidate under the same folds and availability constraints.
4. Track per-experiment MAE, RMSE, model bias, valid prediction count, missingness/failures and calibration/interval coverage where applicable.
5. Rank primarily by **macro-average experiment-level MAE in original target units**, not pooled point-level MAE, to avoid long or densely sampled runs dominating.
6. Prefer a simpler stable model when the nonlinear alternative's improvement is within the uncertainty caused by small group counts.
7. Select champion and challenger without inspecting reserved final holdout; lock config; evaluate once on final groups.
8. Document unfavorable results and any post-hoc adjustment as exploratory rather than a fresh independent test.

Success is not an arbitrary "$R^2 > 0.9$" target. The P0 scientific success criterion is a fair comparison with baselines and clear generalization limits. A measured improvement is desirable but must be earned on held-out groups. If no method consistently beats the baseline, a reproducible negative result still qualifies as honest research, while the product must not advertise accurate soft sensing.

7.6 Evaluation metrics and uncertainty

- **Primary:** macro experiment-level MAE, reported with target units; median and range across experiment groups.
- **Secondary:** RMSE, median absolute error, signed bias, per-mode errors, experiment-level model/baseline ratio, error versus cultivation time, count and percentage of predictions unavailable due to data-quality rules.
- **Uncertainty:** if appropriate, evaluate conformal or residual-based prediction intervals with group-aware calibration partitions; report empirical coverage and width, and acknowledge small-sample dependence. Avoid claiming coverage guarantees unsupported by grouping/exchangeability assumptions.
- **Robustness:** leave-one-experiment-out sensitivity, baseline ablations by sensor family (permittivity/turbidity/gas), sensor dropout and missingness scenarios, early-vs-late cultivation phase errors and batch-vs-fed-batch diagnostics.
- **Negative evidence:** a model that underperforms the baseline for an experiment must be shown, not hidden by an overall average.

7.7 Mechanistic/hybrid extension gate

Gate MH-01: Before fitting a mechanistic model, confirm available, correctly unitized observations for the chosen states and inputs (e.g., substrate, biomass proxy, feed/volume for fed-batch), plus sufficient variation to estimate parameters. If the state equations or parameter identifiability cannot be defended, run a **separately labeled synthetic simulation**, not a data-calibrated biological process model.

Illustrative equations, not assumptions about this specific dataset:

Biomass proxy: $dX/dt = (mu(S) - k_d) * X$
Specific growth: $mu(S) = mu_max * S / (K_s + S)$

For substrate consumption, a growth-associated term might be related to biomass production through a yield coefficient; fed-batch adds feed and dilution terms. These equations require review of the actual cultivation design and measurement availability before use.

Additional science checks: plausible parameter bounds/units; solver convergence; initial-condition uncertainty; local/global parameter sensitivity; multi-start optimization; identifiability profile; out-of-sample error relative to statistical models; transparent distinction between descriptive fit and prospective predictive validity.

Hybrid candidate: $y_hat_hybrid(t) = y_hat_ODE(t) + residual_model(sensor_history_up_to_t)$. Fit any residual corrector on training groups only and evaluate the combined model on untouched experiments. An inferior hybrid result must be reported as such.

8. UX and product specifications

8.1 Dashboard information architecture

Page / tab	Required elements	Critical trust behavior
Overview	Project status, dataset citation, target, experiment counts, model status, research-only notice	No false production or real-time monitoring claims
Data explorer	Select experiment/reactor; plot sensor and offline trajectories; show missingness and sampling times	Distinct symbols for real assays, inline readings and aligned/interpolated values
Model comparison	Baselines and candidates; grouped-CV table; frozen test metrics	Separate development from final holdout results
Prediction review	Measured/predicted overlay, residual plots, warnings for missing/out-of-range data	Never hide unsupported predictions or imply future observations were known
Quality & provenance	Data version, source checksum, units, feature availability, tests, model card	Trace every displayed model output to source and model run
Research limitations	Assumptions, scope, failure cases, open questions, model status	Easily discoverable, not relegated to fine print
Scenarios (P2)	Bounded parameter what-if or mechanistic trajectory plots	Explicitly hypothetical; no process-control recommendation

8.2 Interaction contract

The initial app is **read-only** against stored artifacts. The user may choose filters, experiments, sensor families, model versions and plots. It must not retrain a model silently, alter data, send information externally, or trigger cloud billing. Model calculations requiring recomputation run as an explicit local CLI or controlled user action in a later release.

If a selection lacks measured target values or a model has no supported prediction, the UI must say **“insufficient evidence / unavailable”** and explain why. It must not interpolate a visual curve to conceal missing ground truth.

8.3 Accessibility and performance

- Conventional page hierarchy and descriptive figure titles/axis labels with physical units.
- No reliance on color alone for measured versus estimated series.
- Tooltips or definitions for MAE, RMSE, uncertainty and experiment groups.
- On a standard local development laptop, cached dashboard pages should ordinarily load within a few seconds; measure actual performance before asserting an SLA.
- No external authentication, public account system, paid APIs or live cloud services in V1.

9. System architecture, components and interfaces

9.1 Logical flow

```
Publisher archive / verified checksum
            |
     Immutable local raw data
            |
     Schema + time + unit audit
            |
    Canonical experiment tables
            |
  Frozen experiment-group split plan
            |
     Past-only feature snapshots
            |
     Baselines and model training
            |
  Cross-validation + frozen holdout
            |
    Model artifacts + evidence ledger
            |
   Read-only scientist-facing dashboard
```

9.2 Proposed repository layout

```
biosynvia/
  README.md                 # Working-name / legal caution
  pyproject.toml            # Pinned runtime + tooling
  configs/                  # Target, split and model configs
  data/
    raw/                    # gitignored, license-dependent
    interim/                # gitignored
    processed/              # gitignored
    synthetic_tests/        # safe, generated fixtures
  docs/
    prd/                    # this PRD and decision history
```

```
  scientific_scope.md
  data_dictionary.md
  assumptions_and_limitations.md
  model_card.md
  source_and_license_manifest.md
 src/biosynvia/
  ingestion.py
  schema.py
  units.py
  alignment.py
  features.py
  splits.py
  baselines.py
  train.py
  evaluate.py
  uncertainty.py
  provenance.py
  mechanistic.py          # gated P2
  hybrid.py               # gated P2
 app/streamlit_app.py
 tests/
  unit/
  integration/
  scientific/
 reports/
  experiment_results/
  figures/
  release_checklist.md
 scripts/
  audit_data.py
  build_features.py
  run_experiments.py
  make_report.py
```

9.3 Input/output contracts

- **Ingest CLI:** biosynvia audit --archive <local_path> --manifest <json> verifies provenance and writes inventory/quality reports without modifying raw files.
- **Feature builder:** biosynvia features --target-config <yaml> writes versioned feature snapshots containing experiment/reactor IDs and cutoff times.
- **Training runner:** biosynvia train --experiment-config <yaml> reads a frozen manifest; artifacts include pipeline, config hash, model version, test logs and grouped CV results.
- **Evaluation:** biosynvia evaluate --model-version … --holdout-manifest … produces immutable JSON/CSV metrics and charts; final holdout used only at the signed-off release gate.
- **Dashboard:** reads approved model artifacts and local report files; never mutates reference data.

Actual CLI syntax is a **proposed product API** until implemented. Avoid exposing insecure pickle artifacts to untrusted readers; use an explicit trusted-local-artifact policy, record package versions, and validate file paths.

10. Non-functional requirements and quality gates

ID	Category	Requirement and acceptance evidence
NFR-01	Reproducibility	Clean local install reproduces an approved training/evaluation run within documented numerical tolerance; seeds and versions logged
NFR-02	Traceability	Every published metric links to dataset hash, split IDs, feature config, model config, code revision and experiment count
NFR-03	Data integrity	Immutable raw inputs; source checksum verification; idempotent ingestion and explicit duplicate/missing-file failure reporting
NFR-04	Scientific validity	No mixed-experiment folds; no future-target leakage; all metrics tied to observed assays and correct target units
NFR-05	Input safety	Reject malformed timestamps, unknown units, untracked feature columns and unsupported unseen sensor states
NFR-06	Auditability	Failed tests, inferior model runs and unsupported predictions preserved in a report rather than silently omitted
NFR-07	Privacy	No patient information; no credentials in repository; no external telemetry or data exfiltration by default
NFR-08	Accessibility	Plot units, text equivalents, visible errors and labels distinguish measured/estimated values without color-only encoding
NFR-09	Cost controls	Local-first; no automatic cloud CI or paid services enabled by default; GitHub Actions only with explicit owner authorization
NFR-10	Legal/research identity	Public repository contains no rehosted source archive until rights cleared; BioSynvia working-name warning included
NFR-11	Security	Pinned dependencies, no unsafe processing of untrusted model files, scoped paths and dependency/license inventory

Test pyramid

Unit tests: units, timestamps, schema coercion, negative/duplicate readings, rolling-window cutoffs, deterministic features, split invariants, metric formulas and ODE solver sanity where applicable.

Integration tests: ingest → canonical data → target-at-assay snapshots → grouped folds → model pipeline → metric JSON → read-only dashboard. Use **synthetic fixtures** with intentionally injected leakage and corrupted data.

Scientific regression: paired reactors never split; future perturbation does not alter past predictions; all preprocessing fitted on training groups; raw counts reconcile; frozen holdout cannot be consumed during tuning; results agree when rerun from a clean environment within documented tolerance.

User acceptance: a reviewer can identify target units, measured-versus-estimated points, a failed experiment, the model version, data provenance and intended-use limitations in under five minutes.

11. Decision log and stage gates

Gate	Decision	Pass condition	If fail
G0	Name and dataset use	Working-name warning and source/license record completed	Keep unpublished; resolve with owner/rights holder
G1	Target feasibility	Measured variable, unit, sparse sampling, coverage and inference contract confirmed	Choose defensible alternative target or stop
G2	Data quality	All experiment IDs, reactors, timestamp rules and valid records reconciled	Repair source pipeline; do not train
G3	Leakage safe	Group split and past-only feature tests pass	Block modeling release
G4	Baseline evaluation	Baselines and candidates compared under fixed grouped CV; failures shown	Diagnose; no misleading success claim
G5	Holdout evaluation	Frozen unseen experiments evaluated once with per-group errors	Report outcome honestly; no post-hoc “independent” claim
G6	Presentation V1	Model card, tests, dashboard, reproduction and disclaimers pass	Hold V1 release
G7	Mechanistic eligibility	Observable states, units, model assumptions and identifiability defensible	Keep mechanistic work as labeled synthetic demonstration
G8	Public publication	Brand/trademark review and data redistribution rights addressed	Distribute only internal materials or code with neutral naming
G9	Resume-ready evidence	Applicant personally runs, understands, reproduces and can defend results	Do not list as completed achievement

12. Delivery roadmap, effort and dependencies

Estimates are **focused implementation hours for one developer** familiar with Python/SQL, not a guaranteed completion date. Scientific data issues may materially increase effort. The core V1 is approximately **50–80 hours**; a gated mechanistic/hybrid extension adds approximately **18–35 hours**.

Phase	Effort	Implementation work	Sign-off artifact
P0 — Initiation	3–5 h	Repository, scope, definitions, source and branding/legal record	Project charter; research hypothesis; setup works
P1 — Dataset audit	8–12 h	Download/checksum, inventory, units, missingness, paired experiment analysis, target audit	Data dictionary + target selection gate G1
P2 — Canonical and features	8–13 h	Normalize tables, compare alignment, timestamp rules, past-only features, group splits	Feature pipeline + leakage tests G2/G3
P3 — Statistical modeling	10–16 h	Baselines, Ridge/PLS and nonlinear model, group CV, tracked experiments	Benchmark report G4
P4 — Independent evaluation	11–18 h	Frozen holdout, per-group metrics, subgroup analysis, ablations, uncertainty caveats	Scientific results and model card G5
P5 — Dashboard & package	10–16 h	Streamlit UI, clean run, test harness, README and presentation	Demonstrable V1 G6
P6 — Advanced extension	+18–35 h	Optional mechanistic/ODE + hybrid modeling, sensitivity, identifiability, scenarios	Gated research appendix G7

Execution recommendation: prioritize **G1 → G3 → G5** over branding or elaborate UI. Present a credible intermediate package as soon as P4 passes: one experiment-held-out benchmark, one failure analysis and a precise limitations narrative. The advanced hybrid extension may be deferred without making V1 misleading.

Possible calendar pacing

- At **15 focused hours/week**, V1 typically spans roughly **4–6 weeks**.
- At **20–25 focused hours/week**, V1 typically spans roughly **2–4 weeks**.
- An extra **18–35 hours** applies only if mechanistic/hybrid work passes its data gate.

These are planning ranges, not promises about future work or scientific performance.

13. Milestone acceptance matrix and definition of done

13.1 Data-ready (M1)

- Source ZIP checksum verified; archive structure reconciled to 12 experiments/24 reactors or discrepancies recorded.
- All prediction-relevant fields have source/units and documented quality policy.
- Batch/fed-batch distribution and target coverage reported by experiment.
- A valid measured target is selected, or an explicit no-go decision is documented.

13.2 Model-ready (M2)

- Paired reactors share partitions; raw fold manifests version-controlled.
- Past-only features tested; preprocessing leakage tests pass.
- Naive baseline and candidate model protocols fixed before final evaluation.
- Every metric references target units, group IDs and sample counts.

13.3 Research-ready (M3)

- A frozen evaluation reports macro experiment-level MAE, secondary metrics, model/baseline comparison and failures.
- At least one feature-family ablation and a missing-sensor robustness check.
- Negative, null, unstable or mode-specific results openly described.
- Reproduction succeeds on a clean environment and model card is attached.

13.4 Demo-ready (M4)

- A local reviewer can select an experiment and see measured/predicted trajectories, missingness and unsupported predictions.
- UI identifies input variables, data provenance and model version.
- A concise technical presentation includes problem, dataset, baseline, validation, results and limitations.
- All research-only disclaimers and branding/data rights gates have been respected.

13.5 Advanced-ready (M5, optional)

- Mechanistic assumptions reviewed, parameters plausibly identifiable and solver tests pass; or clearly labeled synthetic-only exploration.
- Hybrid method compared on identical held-out groups, regardless of whether it beats statistical baseline.
- Scenario examples explicitly flagged as hypothetical, not validated process optimization.

14. Risk register and contingency plan

Risk	Severity	Signal / early detection	Response
Trademark confusion with SYNVIA	**High**	Similar mark in scientific/software-related classes	Working title only; avoid public brand launch; legal clearance/name change gate
Dataset raw redistribution rights unclear	**High**	Zenodo Rights license field blank	Rehost no raw files; request clarification; cite original DOI

Risk	Severity	Signal / early detection	Response
Target absent or poorly covered	**High**	Offline field has low valid count or inconsistent units	Select alternate scientifically valid target or stop
Leakage from paired experiments	**High**	Same experiment_id appears on both sides	Automated hard-fail test
Time/target leakage from alignment	**High**	Future readings influence earlier features or labels filled across time	Cutoff-based builders, injection tests, no evaluation of interpolated labels
Small number of independent groups	**High**	High fold variance, mode imbalance	Macro per-experiment metrics, LOEO sensitivity, cautious uncertainty
Mechanistic unidentifiability	**High**	Unstable fit, multiple equivalent parameter sets	Do not claim data-backed mechanistic model; synthetic-only extension
False “Digital Twin” impression	**High**	Dashboard or resume overstates prototype	Mandatory disclaimers, claim review
No model beats baseline	Medium	Negative group-held-out deltas	Report negative result; identify conditions and limitations; avoid performance claims
Sensor availability / missingness	Medium	Dropouts or unusable signal ranges	Missingness flags, ablations, abstention policy
Development scope creep	Medium	Dashboard/advanced models before benchmark	Phase gates; freeze V1; defer P2 features
Unapproved cloud charges	Medium	Automatic workflows or hosted compute	Local-only by default; no unattended CI or paid services without approval
Reproducibility failures	Medium	Different outcomes across environments	Dependency lock, deterministic configs, seed/environment capture

15. Research outputs, technical storytelling and portfolio ethics

15.1 Required reviewer package

1. README.md with problem, source, setup, reproducibility command and limitations.
2. Data dictionary, target-selection decision, manifest and license/attribution statement.
3. Source→features→split→model lineage diagram and tests.
4. Baseline/model comparison with per-experiment errors and honest negatives.
5. Measured-versus-predicted plots, missing-data diagnostics, and uncertainty/applicability assessment.
6. Model card and experiment registry with configurations and exact commands.
7. Local dashboard demonstration with no hidden data or commercial claims.
8. Six-slide technical presentation plus a 2-minute and a 10-minute narrative.

15.2 Hiring-manager presentation outline

Slide	Content	What it demonstrates
1	Scientific problem and intended user decision	Can translate a process question into a modeling task
2	Dataset and experimental design	Understands source data and biological independence
3	Preprocessing, leakage controls and baselines	Engineering and scientific rigor
4	Model comparison and holdout protocol	Correct validation and evidence-based method selection
5	Results **including failures and uncertainty**	Sound model interpretation and scientific honesty
6	Dashboard, limits and next scientific experiment	Usable outputs without overstating readiness

15.3 Resume claim policy

Before running this project: it is an *implementation plan*, not an accomplishment. **After personally reproducing and validating V1:** a truthful project description may mention a research prototype, real CHO cultivation data, methods actually implemented, cross-experiment validation, and actual measured error metrics. **Never** describe a deployed Digital Twin, CGT process manufacturing experience, GxP validation, regulatory approval, model-driven optimization gains, or professional senior scientist experience unless independent evidence exists.

Example wording is **not to be used until verified**: “Developed a research-grade CHO soft-sensor prototype using inline/offline cultivation measurements; benchmarked statistical and ML models with experiment-group holdouts and documented uncertainty and failure modes.” Results, model names and numbers must be inserted **only after successful personal execution and evidence review**.

16. Open questions and owner decisions

ID	Question	Responsible	Decision deadline
OQ-01	Which offline variable has adequate independent-group coverage, units and scientific usefulness?	Developer + scientific reviewer	End of P1
OQ-02	Which inline signals are available in a realistic at-time inference scenario?	Developer	End of P1
OQ-03	Can the repository license be independently verified, and are raw files redistributable?	Developer/rights holder	Before public release
OQ-04	Which two experiment IDs, if any, should form the frozen balanced holdout?	Developer	Before model tuning

ID	Question	Responsible	Decision deadline
OQ-05	What is a meaningful error tolerance in the original target units and intended use?	Scientific reviewer	Before G5 claim of utility
OQ-06	Are substrate, feed and volume variables sufficient to calibrate any mechanistic model?	Developer + process expert	Before P6
OQ-07	Do uncertainty intervals have acceptable empirical group-held-out coverage?	Developer	During P4
OQ-08	Can BioSynvia be safely used beyond an internal working title?	Trademark professional / owner	Before public branding
OQ-09	What J&J modeling problem would the hiring manager consider most representative?	Owner, via lawful referral conversation	As early as feasible

17. First implementation sprint — executable action plan

The first sprint should produce a **go/no-go scientific data audit**, not a flashy model. The following are ordered actions for the developer, not claims of completed work.

1. Create a private repository under a neutral, renameable working path (`bioprocess-softsensor` if preferred) with Python environment, tests and an explicit no-cloud-cost policy.
2. Download the archive through the publisher; log DOI, retrieved filename, byte size, checksum and the still-unresolved license status.
3. Enumerate all .xlsx and aligned CSV files; parse EXP/reactor/mode IDs from file names; verify the expected 12×2×2 aligned file structure.
4. Generate machine-readable dictionaries of original and aligned columns, types, units, timestamps, invalid values and missingness by reactor.
5. Plot one representative batch and one fed-batch run with assay time markers and sensor trajectories; document alignment caveats.
6. Evaluate viable cell concentration/density and any other measured offline target against target-selection criteria; approve a target or stop.
7. Write `inference_contract.md`: sensor-only versus lab-assisted, cutoff time and any reporting delay assumptions.
8. Freeze proposed group split IDs after reviewing experiment metadata, without looking at final test model performance.
9. Implement the naive baseline and simple unit tests for group separation and past-only feature generation.
10. Review the audit pack: decide whether to proceed with model development, change target, or revise scope.

Sprint output: source manifest, data dictionary, quality report, reproducible EDA notebook, target decision, split design, first baseline, and a short transparent scientific status report. This is the earliest useful material to show a scientific reviewer.

18. Source register and attribution

Source facts and **design recommendations** have been kept distinct throughout the PRD. These references support source metadata, legal risk context and the validity of recommended methodology; they do **not** verify any BioSynvia implementation or model result.

- **[S1]** Uhlendorff, S., Fulek, R., Eimler, J., Pein-Hackelbusch, M. and Frahm, B. (2026). *Dataset Based on Chinese Hamster Ovary (CHO) Cultivations including Turbidity, Permittivity, O2 and CO2 Measurements.* Zenodo, DOI: **10.5281/zenodo.20829178.** https://zenodo.org/records/20829178
- **[S2]** Uhlendorff et al. (2026). *Data Descriptor: Multimodal Dataset of Bioreactor Cell Cultivations Including Turbidity, Permittivity, O2 and CO2 Measurements*, non-peer-reviewed preprint posted November 4, 2026. https://www.preprints.org/manuscript/202611.0377 . Article CC BY 4.0 statement **does not establish the archive's license**.
- **[S3]** U.S. Food and Drug Administration. *PAT — A Framework for Innovative Pharmaceutical Development, Manufacturing, and Quality Assurance*. Guidance for Industry, October 2004. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/pat-framework-innovative-pharmaceutical-development-manufacturing-and-quality-assurance
- **[S4]** scikit-learn. *LeaveOneGroupOut* and grouped cross-validation documentation. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.LeaveOneGroupOut.html
- **[S5]** U.S. Patent and Trademark Office. *Likelihood of confusion*. https://www.uspto.gov/trademarks/search/likelihood-confusion
- **[S6]** SYNVIA trademark record, reported Reg. No. **7505668**, Serial **98083883**, scientific research/software-related services. Third-party record: https://furm.com/trademarks/synvia-98083883 . For authoritative current status, verify in USPTO TSDR/trademark search.
- **[S7]** Synvia operating company, pharmaceutical/clinical research and digital solutions. https://synvia.com/en/
- **[S8]** User-provided July 2026 *Johnson & Johnson Innovative Medicine — Senior Scientist/Modeler* job description. Used for role alignment only; its current live/open application status is not established here.

Document decision: BioSynvia is a viable **internal research plan**, contingent on the target-data and naming/redistribution gates. This PRD authorizes planning, not claims of a finished scientific model or a legally cleared public biotech brand.